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Artificial intelligence and recommender systems in e-commerce. Trends and research agenda

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ABSTRACT

Combining recommendation systems and AI in e-commerce can improve the user experience and decision-making. This study uses a method called bibliometrics to look at how these systems and artificial intelligence are changing. Of the 120 documents, 91 were analyzed. This shows a growth of 97.16% in the topic. The most influential authors were Paraschakis and Nilsson, with three publications and 43 citations. The magazine Electronic Commerce Research has four publications and 60 citations. China is the top country for citations, with 120, followed by India with 25 publications. The results show that research increased in 2021 and 2022. This shows a shift towards sentiment analysis and convolutional neural networks. The identification of new keywords, such as content-based image retrieval and knowledge graph, shows promising areas for future research. This study provides a solid foundation for future research in e-commerce recommender systems.

1. Introduction

The combination of recommender systems and AI in e-commerce has led to a lot of research. There are different types of recommendations, including virtual assistants and online suggestions made in real time. These systems use AI to make e-commerce better for users and help them make decisions (Reyes, 2023). They analyze information to provide personalized recommendations.

The growing availability of data and the complexity of user behavior patterns has led to an increased demand for personalized recommendations. To improve this, AI techniques have been added to recommender systems to make predictions more accurate and deal with problems like not having enough data and starting from scratch. Also, the many products on e-commerce sites can confuse customers, making it hard for them to find the best item.

Recent research has explored AI techniques to address specific challenges in recommender systems. For example, a recommender system based on deep neural networks has been proposed to address the recommendation challenge in cold start cases on e-commerce platforms. A deep neural network model was developed to suggest fashion combinations using additional e-commerce information (S. Wang & Qiu, 2021). A hierarchical recommendation system was introduced that integrates online user opinions to improve e-commerce recommendations (Islek & Oguducu, 2022).

Also, researchers have studied ways to combine different AI techniques to improve recommendation systems for e-commerce. One

Abbreviations: CNNs, Convolutional Neural Networks; CNN, Convolutional Neural Network; GRU, Gated Recurrent Units; RNN, Recurrent Neural Network; RNNs, Recurrent Neural Networks; WSDM, Web Search and Data Mining; SDAE, Sparse Denoising Autoencoder; LSTM, Long Short-Term Memory; IEEE, Institute of Electrical and Electronics Engineers; AI, Artificial Intelligence; Variables, Adjustable Control Parameters; Number of publications, Search equations; Number of citations, Inclusion/exclusion criteria; Temporal patterns of keyword usage, Selection thresholds; Keyword co-occurrence, Bibliometric indicators; Frequency of keyword usage, Analysis tools (Microsoft Excel®, VOSviewer®); Validity of keyword usage, Data validation process.

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example is a customer-focused system that uses a mix of sentiment analysis (Karn et al., 2023). This means using AI to identify, extract, and analyze emotions, opinions, and attitudes expressed in online text (Bawack, Wamba, Carillo & Akter, 2022; He & Liu, 2024). An unsupervised machine learning method has been used to develop an image recommendation system, which is similar to those used in e-commerce. This topic has generated a lot of interest in the research community. These systems play an important role in improving user experience and decision making in online business environments. Online recommender systems are important and we study how they are used in e-commerce. They help make personalised recommendations (Cabrera-Sánchez, Ramos-de-Luna, Carvajal-Trujillo & Villarejo-Ramos, 2020). Despite recent recommendation systems being good at making recommendations, they have some problems, including being able to handle lots of data, starting from scratch, and having lots of data (Roy & Dutta, 2022). The article also shows how artificial intelligence can help find natural communities in complex networks, which is important for e-commerce (Chattopadhyay, Basu, Das, Ghosh & Murthy, 2021). These studies show how recommender systems and artificial intelligence help e-commerce.

Anitha and Kalaiarasu (2021) show how collaborative machine learning can improve e-commerce recommender systems. Their research shows how optimizing these methods helps provide more accurate and personalized suggestions. The study also addresses how users interact with e-commerce platforms (H. Wang et al., 2019) and proposes a deep neural network-based recommendation system to address the cold start problem (H. Wang et al., 2019).

Artificial intelligence affects how features are chosen in recommender systems. The study highlights the importance of feature selection based on conditional entropy in automated e-commerce recommender systems (Sharma & Sadagopan, 2022). This study shows how AI can improve e-commerce recommender systems.

In e-commerce, intelligent search algorithms have improved efficiency and effectiveness. The gravitational search algorithm is one of these systems that has proven useful in optimizing complex problems. This approach improves online recommendations by optimizing threshold settings. This improves user experience and recommendation accuracy. It has also been used to identify dynamic systems in the context of electronic commerce. This method, inspired by classical mechanics, helps improve recommendation systems. It makes recommendations more adaptable and personal to users (Mohammadi & Zahiri, 2018). Integrating optimization into recommender systems makes them more adaptable and accurate.

Particle swarm optimization can solve many types of problems. It can help e-commerce systems work better. It can also help users get better recommendations. Langs (2012) says this is a good approach.

The effectiveness of using the inclined plane system algorithm to solve problems with multiple local optima was tested. The study shows how inclined plane-based optimization can help with complex e-commerce challenges. They can also adapt recommendation strategies to suit different users in online shops (Darband, Mahboob, Shahbandegan & Razavi, 2020). Integrating these advancements into intelligent optimization algorithms can enhance the adaptability and accuracy of recommendations, which can improve customer satisfaction and loyalty in e-commerce.

Despite recent advances in recommender systems and AI in e-commerce, there is still much to learn. A model has been created to predict how useful reviews are on e-commerce platforms. However, there is still more to be done to improve the accuracy of recommendations (Almutairi & Abdullah, 2020). Although a hierarchical recommendation system based on online user reviews has been introduced, it is essential to investigate how these techniques could be adapted and refined to deal with the increasing complexity and diversity of data in e-commerce.

Also, while product ad recommendation in e-commerce has been addressed using deep learning and distributed rendering techniques. However, it is important to investigate how these strategies could be used more widely across different types of products and users (Zhou, 2020).

This study will help e-commerce and AI by combining and expanding existing knowledge. The study suggests using machine learning to understand how to use machine learning in e-commerce. The study shows how to create effective recommendation systems. It explains how to use advanced algorithms to improve the user experience and make recommendations more effective (Loukili, Messaoudi & El Ghazi, 2023). Furthermore, the study provides a comprehensive overview of AI-driven recommendations in e-commerce research (Necula & Păvăloaia, 2023). The study aims to show how existing knowledge on recommendation systems for e-commerce can be improved. By understanding and addressing the problems found by other researchers, the study can help improve and innovate. A study on online recommender systems by Cabrera-Sánchez et al. (2020) looks at why they are used in e-commerce. Combining this with the proposed study could help identify new research areas and focus points to improve recommender systems in online commercial environments. Similarly, an examination of AI in recommendation systems provides an overview of the most recent trends and advancements in the field (Zhang, Lu & Jin, 2021). By combining this study with the proposed research, we can gain a more comprehensive understanding of how AI techniques are transforming e-commerce recommendation systems and how they can be effectively applied to enhance user experience and increase recommendation efficiency.

This article says that new ideas for e-commerce recommendations and AI are needed to meet the growing demands and expectations. This approach is needed to overcome current research limitations and challenges. Although there has been progress, there is still more to do. For example, Acharya, Sassenberg and Soar (2023) say that we need to understand how cognitive absorption affects people's use of AI-based recommender systems. Sahu and Gaur (2024) say that recommendation systems in e-commerce should be more understandable. These studies show that we can understand and use AI in e-commerce better. This study is important because of this.

A recent study by Shin et al. (2021) presents a new way to represent users in recommender systems for e-commerce. This study can help us understand how to make recommendations more personal and relevant to users. This will help us improve e-commerce strategies based on personalized recommendations.

This research looks at how AI recommender systems are used in e-commerce. It aims to set out a research agenda for future work. To this end, we propose the following research questions:

When did interest in recommender systems and AI techniques in e-commerce peak?
How many scientific articles on recommender systems and AI in e-commerce have been published?
What are the main research papers on recommender systems and AI in e-commerce?
What has changed in the topics studied in scientific articles on recommender systems and AI techniques in e-commerce?
What are the main topics on recommender systems and AI techniques in e-commerce?
What are the new and growing keywords in recommender systems and AI techniques in e-commerce?
What topics should be included in a research agenda on recommender systems and AI techniques in e-commerce?

The article starts with a look at how recommender systems and AI techniques are used in e-commerce. The Methodology section explains how the study was done and how the data was collected and analyzed. It answers the questions above. The results are in the Results section. Finally, the article explains what the results mean.

1.1. Motivations

This section looks at why researchers are studying recommender

systems and AI techniques in e-commerce. The reasons include the increasing use of e-commerce platforms, the need for personalised recommendations to improve the user experience, and the potential of AI to improve recommendation strategies. It also discusses the gaps and challenges in existing research that motivate further research in this field.

1.2. Innovations

This article looks at new ways of doing research on recommendation systems and AI in e-commerce. It talks about new types of recommendation systems and algorithms that combine different AI techniques. It also discusses new ways of making recommendations more accurate and efficient. Finally, it looks at how AI can be used to solve specific problems, such as problems with starting new sessions and user interaction.

1.3. Contributions

This study assesses the impact of research in the field of electronic commerce and recommender systems. It consolidates existing knowledge, expands research horizons, and addresses gaps in the existing literature. Previous studies have contributed insights from bibliometric analysis, recommendations for future research, and implications for practitioners and stakeholders in the e-commerce industry. The proposed study contributes to academic knowledge by explaining how AI-powered recommender systems work in e-commerce. The article is structured as follows:

1.4. Overview of article sections

The article includes the Abstract, which is a concise summary of the research objec-tives, methods, findings, and conclusions. It also includes the Introduction, which sets the context for the study, outlines the research problem, objectives, and significance, and provides an overview of the research approach. The methodology describes the research design, data collection methods, analytical techniques, and any tools or soft-ware used for analysis. The results section presents the key findings of the study, in-cluding quantitative and qualitative data, visualizations, and statistical analyses. The discussion section interprets the results in the context of existing literature, discussing implications, limitations, and future research directions, and providing insights into the broader significance of the findings. Finally, the conclusions summarize the main findings of the study, highlight contributions to the field, and offer concluding remarks on the research implications and potential impact.

2. Materials and methods

This research follows the PRISMA-2020 guidelines, as presented in Page et al. (2021). It looks at how recommender systems and AI techniques have changed over time and what they are used for in e-commerce. This approach will help us understand what has been done in this field and what still needs to be done. By using a good method, we can give a good picture of what we know about this changing field.

2.1. Eigibility criteria

This bibliometric review focuses on recommender systems and AI techniques in e-commerce. To select sources, we look for records with titles and keywords related to the subject. Second, records are included that address the intersection of recommender systems and AI techniques with e-commerce. This inclusive approach ensures a complete and accurate representation of relevant research in this area.

The exclusion process has three steps. The first is to remove records with indexing errors. The second is to remove documents with missing text. This only applies to systematic literature reviews. The third is to remove incomplete records. Finally, the third phase will focus on eliminating documents with incomplete indexing information. This ensures the reliability and coherence of the sources submitted to the bibliometric analysis.

2.2. Source of information

This study used Scopus and Web of Science as its main sources of data. These databases are the best for scientific and academic research because they cover a lot of relevant literature and have good analytical tools. Previous research has shown that Scopus and Web of Science are reliable and representative sources for this bibliometric analysis (Zhu & Liu, 2020).

2.3. Search strategy

Two search equations were developed to refine the search for information in the Scopus and Web of Science databases. These equations were designed with the criteria for bibliometrics and the structure of the selected databases in mind. Specialized search equations are needed to find studies on recommender systems and AI in e-commerce.

For the Scopus database: (TITLE ("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network algorithms" OR "natural language processing" OR "computer vision techniques" OR "reinforcement learning") AND TITLE (e-commerce OR "electronic commerce" OR ecommerce OR "online commerce" OR "online store*" OR "Internet commerce" OR "Internet store*" OR "virtual commerce" OR "virtual shop*" OR "cyber commerce" OR "online commerce" OR "web shop*") AND TITLE ("recommendation system*" OR "recommendation engines" OR "personalization system*")) OR (KEY ("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network algorithms" OR "natural language processing" OR "computer vision" OR "reinforcement learning") AND TITLE (e-commerce OR "electronic commerce" OR ecommerce OR "online commerce" OR "online store*" OR "Internet commerce" OR "Internet store*" OR "virtual commerce" OR "virtual shop*" OR "cyber commerce" OR "online commerce" OR "web shop*") AND KEY ("recommendation* system*" OR "recommendation* engines" OR "personalization system*"))

For the Web of Science database: (TI=("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network algorithms" OR "natural language processing" OR "computer vision" OR "reinforcement learning") AND TI=(e-commerce OR "electronic commerce" OR ecommerce OR "online commerce" OR "online store*" OR "Internet commerce" OR "Internet store*" OR "virtual commerce" OR "virtual shop*" OR "cyber commerce" OR "online commerce" OR "web shop*") AND TI=("recommendation system*" OR "recommendation engines" OR "personalization system*")) OR (AK=("artificial intelligence" OR "deep learning" OR "machine learning" OR "neural network algorithms" OR "natural language processing" OR "computer vision techniques" OR "reinforcement learning") AND TI=(e-commerce OR "electronic commerce" OR ecommerce OR "online commerce" OR "online store*" OR "Internet commerce" OR "Internet store*" OR "virtual commerce" OR "virtual shop*" OR "cyber commerce" OR "online commerce" OR "web shop*") AND AK=("recommendation* system*" OR "recommendation* engines" OR "personalization system*"))

2.4. Data management

This bibliometric study used Microsoft Excel® for Microsoft 365 MSO (version 2403 build 16.0.17425.20124) 64-bit to extract, store, and process information from selected databases. Microsoft Excel® made it easy to manage and analyze the data. The free program VOSviewer® in its latest version 1.6.20 for Microsoft Windows Systems, updated on October 31, 2023, was also used. This tool let us analyze how terms appear together and create graphics showing this. This is the same approach used in previous research (Kuzior & Sira, 2022). Microsoft Excel® and VOSviewer® helped us understand patterns and

relationships in the scientific literature.

Microsoft Excel® is a good tool for data visualization. It has been discussed in the literature (Raubenheimer, 2017) and is good for data representation and analysis, particularly in the context of bibliometrics. Kuzior and Sira (2022) used VOSviewer® to analyze research on blockchain technology. This supports using it to represent relationships and patterns in the scientific literature related to recommender systems and AI techniques in e-commerce.

2.5. Selection process

Page et al. (2021) said it's important to use internal automated classifiers to find relevant studies and to check that the studies are being classified correctly. This study used Microsoft Excel® to do this. The study investigators developed these tools and each team member used them to assess inclusion and exclusion criteria. This approach was used to ensure that no relevant studies were overlooked or misclassified. By combining and validating the results, the selection process was made more accurate and consistent.

2.6. Data collection process

The Page et al. (2021) guidelines say that we must describe how we collected data from bibliographic reports. This means saying how many reviewers worked on each report, whether they worked alone, and how we checked the data with the study investigators. This bibliometric research looked at recommender systems and AI techniques in e-commerce. Microsoft Excel was used to get data from the two databases. All authors checked the data and did their work separately. They checked the data together to make sure it was right. The results were checked together to make sure they were reliable and consistent.

2.7. Data elements

This study collected results from articles that matched the research goal. We used search equations in each database to find these articles. If we found unclear or incomplete information, we excluded it. This was to keep the results consistent and to ensure the integrity of the results.

2.8. Assessment of the risk of bias of the study

This study collected results from articles that matched the research goal. We used search equations in each database to find these articles. If we found unclear or incomplete information, we excluded it. This was to keep the results consistent and to ensure the integrity of the results.

2.9. Impact measures

The specification of effect measures is distinct in this research. It is based on secondary sources such as scientific articles and metadata. Traditional impact measures are not applied in the same manner. Instead, bibliometric indicators are used. The results are presented in terms of the number of publications, the number of citations, and the pattern of temporal use of each keyword. Furthermore, VOSviewer® is used to analyze existing nodes and identify thematic associations.

2.10. Synthesis methods

This process explains how studies are chosen for the synthesis. It also explains how the data is prepared and the results are presented. For this, bibliometric indicators were used to assess quantity, quality, and structure. These indicators follow the guidelines for scientific publications (Durieux & Gevenois, 2010). These indicators were automatically applied to documents that passed the three exclusion stages. This approach allowed for an unbiased assessment of the quantity and quality of publications, as well as the identification of structural patterns. This contributed to the accurate and quantitative visualization of the results.

2.11. Assessment of reporting bias

This paper looks at how a lack of results in the synthesis could lead to bias. This bias could be due to reporting bias. Some synonyms in the IEEE Thesaurus might affect the study. Also, texts with incomplete indexing may omit valuable information. These potential biases are considered when interpreting and applying bibliometric results. They affect how representative and complete the studies analyzed are.

2.12. Assessment of certainty

This review looks at how sure we can be about the evidence. This review looks at the whole study, not just one part. It looks at things like how to choose which studies to include and exclude, how to measure things like how many times a study has been published, and how to spot any problems with the study. These are explained in the method. The review also looks at how sure we can be about the results, points out any problems with the study, and explains why the study is important (Fig. 1).

This bibliometric study involved selecting and excluding records. First, records were found in each source, then duplicates were removed. Then, three exclusion stages were applied, reflecting a comprehensive screening process. After these stages, 91 articles were included. The articles in this study are the basis for a full examination of recommender systems and AI techniques in e-commerce. This study's methodology is based on a systematic and rigorous approach to data collection and analysis.

3. Results

This study shows important research trends. This figure shows a big increase in scientific production, reflecting growing interest in the relationship between recommender systems and artificial intelligence strategies in the context of electronic commerce. The most articles on this topic were published in 2020, 2021, 2022, and 2023. This is an excellent trend in recent research.

The analysis of the main authors in recommender systems and AI techniques in e-commerce shows different approaches and research strategies. It shows that being productive is important and that you can make a big impact if you focus your work. Two groups were identified, as shown in Fig. 3. The first group includes Paraschakis and Nilsson, who have made a big impact with their work. These studies have made a big impact on the literature. The second group of authors is distinguished by a high impact in terms of scientific productivity, although they have a relatively low index. Authors like Zhou L., Friedman D., Tan F., Zhang Y., and Zhao Q. have made important contributions to research.

Fig. 4 shows that the most important journals in terms of citations can be divided into three groups. The first group includes journals like Electronic Commerce Research, which are very productive and impactful. The second group includes journals with low scientific productivity but high impact. Examples include Recsys 2015-Proceedings Of The 9th Acm Conference On Recommender Systems and Journal Of Visual Communica-tion And Image Representation. The third group is distinguished by high scientific productivity, but less impact in terms of citations. Applied Soft Computing is a good example of a journal in this category.

A detailed analysis of countries shows three groups (Fig. 5). The first group includes China, India, and the United States, which are major players in science. The second group includes countries like Sweden and Spain, which have a moderate scientific productivity index but a high impact on the subject. This shows that quality is more important than quantity. The third group is distinguished by high scientific productivity, but not necessarily by the number of citations.

Fig. 6 shows how the literature on recommender systems and AI

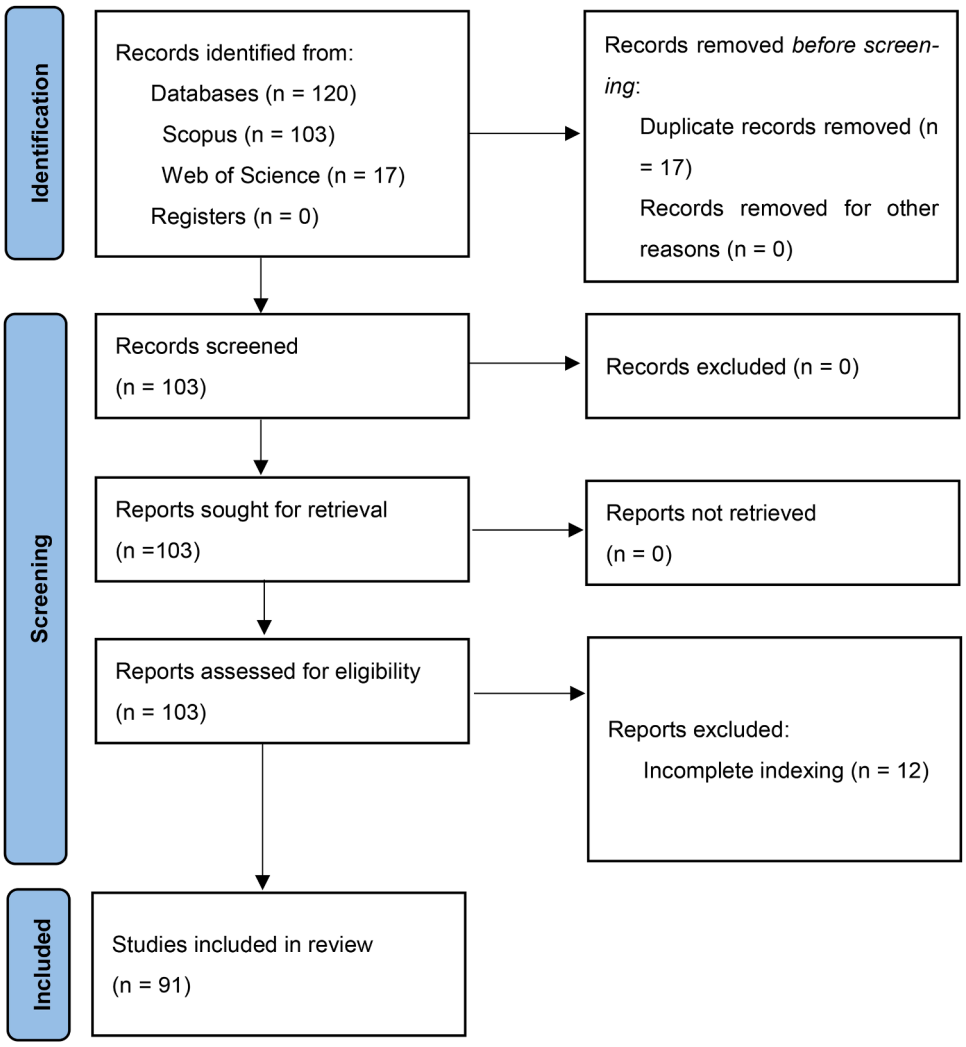

Fig. 1. PRISMA flowchart. Own elaboration from Scopus and Web of Science.

techniques in e-commerce has changed over time. This analysis shows the most common keywords in each year of the study, from 2007 to 2023. The year 2007 marks the start of the research, when multiagent systems emerged. In recent years, there has been a lot of research on topics like sentiment analysis, convolutional neural networks, collaborative filtering, session-based recommendation, and implicit feedback. This research provides a more complete picture of the field.

Fig. 7 shows the main keyword co-occurrence network, which has six thematic clusters. These are shown in different colors, with yellow being the most prominent. This cluster includes terms like collaborative filtering, sparse rating, implicit feedback, and recommendation, which show its importance in the field. The red cluster is about sentiment analysis, neural networks, accuracy, and Bert. Other clusters can be identified in light blue, purple, green, and blue. These demonstrate how the keywords are related to recommendation systems and artificial intelligence techniques in electronic commerce.

Fig. 8 shows how keywords are used in recommender systems and e-commerce. This Cartesian plane has two axes. The X-axis shows how often keywords are used, and the Y-axis shows how useful they are over time. This makes four different quadrants.

In Quadrant 2 are terms that are rarely used but are important. These include terms like content-based image retrieval, content-based filtering, convolutional neural network, and knowledge graph. In Quadrant 1, we find consolidated and emerging concepts. These are used a lot and are always interesting. Examples include sentiment analysis and collaborative filtering.

4. Discussion

The results are based on a methodical approach, following the PRISMA-2020 guidelines. This methodology provides a structured framework for a comprehensive bibliometric study, allowing the identification and synthesis of relevant information in a transparent and reproducible manner. The methodology ensures that the data is searched, selected, and analyzed correctly, which makes the results more valid and reliable.

The discussion section of this report explains the results in detail. The findings are presented in detail, allowing a deep understanding of the trends, how keywords interact, and how the field of recommender systems and artificial intelligence techniques in electronic commerce has evolved. Also, we discuss the practical results and the study's limitations. Also, the keywords are grouped by function, the biggest gaps in current research are found, and a plan for future research is made.

4.1. Analysis of the growth of scientific literature on recommender systems and ai techniques in e-commerce

A lot of scientific research was done in the fields of recommender systems and artificial intelligence techniques applied to e-commerce between 2020 and 2023 (see Fig. 2). In 2020, a study showed how to

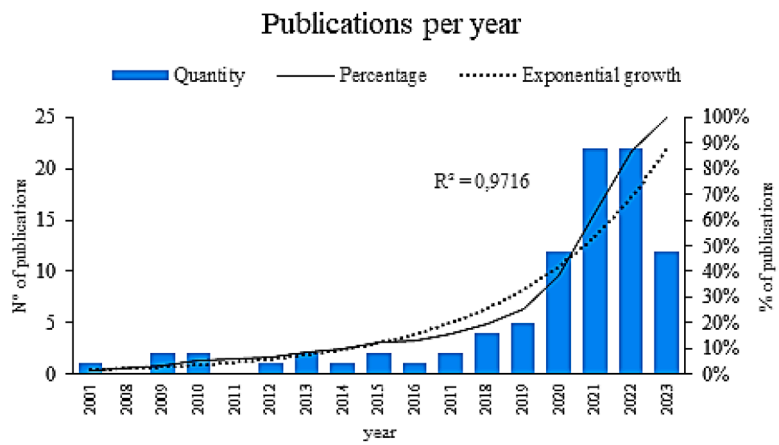

Fig. 2. Publications per year. Compiled from Scopus and Web of Science.

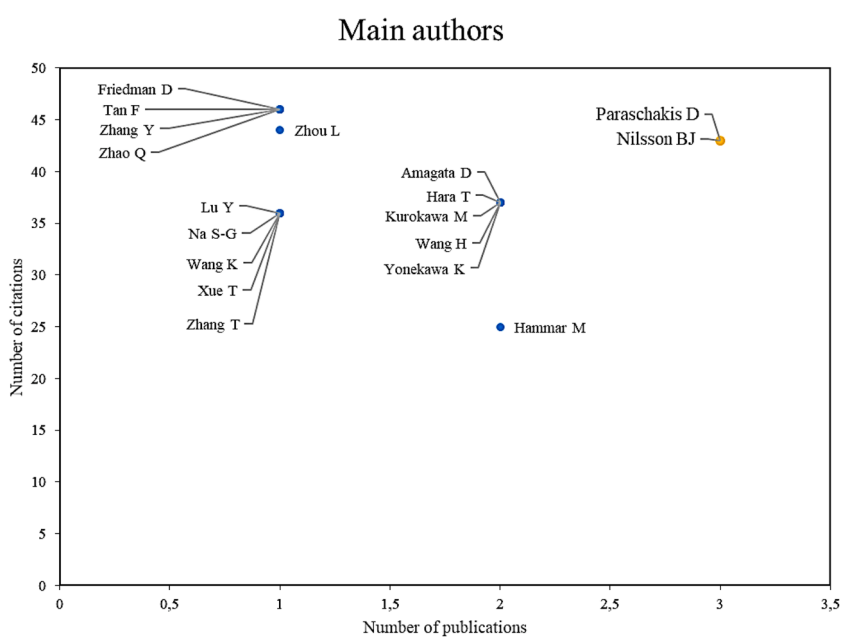

Fig. 3. Main authors. Compiled from Scopus and Web of Science.

make personalized recommendations in e-commerce using deep learning clustering (S. Wang & Qiu, 2021). In 2022, a study was done to deal with sparse score matrices in e-commerce recommender systems. The study proposed a hybrid deep learning approach based on LSTM, SDAE, and latent factor (Pujastuti, Laksito, Hardi, Perwira & Arfriandi, 2022). Another study from that year created a hierarchical e-commerce recommendation system based on online users' opinions (Islek & Oguducu, 2022).

In 2021, a machine learning-based e-commerce recommendation system was introduced (Tahir, Enam & Mustafa, 2021). A deep neural network model was developed for e-commerce fashion recommendations using additional information (S. Wang & Qiu, 2021). Finally, in 2023, various authors developed a hybrid recommendation system for e-commerce that focuses on the customer and uses sentiment analysis for greater effectiveness (Karn et al., 2023).

Researchers are interested in developing and improving recommender systems and AI techniques for e-commerce. These systems address challenges in an evolving field.

4.2. Analysis of research references on recommender systems and AI techniques systems and AI techniques in e-commerce

Researchers Paraschakis D and Nilsson BJ have shown they are leaders in recommender systems and AI techniques in e-commerce. In one of their papers, they compared the best recommender systems in an industrial context. This helped them understand how well recommender systems work in e-commerce (Paraschakis, Nilsson & Holländer, 2015). A few years later, these authors conducted an experiment using Thompson sampling to generate recommendations, presenting new approaches in e-commerce (Brodén, Hammar, Nilsson & Paraschakis, 2018).

He also worked with researcher Zhou L on a recommendation system for e-commerce platforms. This is an innovative way to improve advertising recommendations (Zhou, 2020).

Zhao Q and other researchers also looked at e-commerce recommendations with personalized promotions. They came up with a way to combine product recommendations and promotion strategies for each user. This approach improves user experience and sales in e-commerce (Zhao, Zhang, Friedman & Tan, 2015). Together, these studies show how authors influence recommender systems and AI techniques in

e-commerce.

The following journals have helped improve research in recommendation systems and AI techniques applied to e-commerce: Electronic Commerce Research, Recsys 2015-Proceedings Of The 9th Acm Conference On Recommender Systems, Journal Of Visual Communication And Image Representation, and Applied Soft Computing. (See Fig. 4.) The journal Electronic Commerce Research has articles about advertising recommendations that help identify communities in complex networks. These include Zhou (2020) and Chattopadhyay et al. (2021). The 2015 ACM Conference on Recommender Systems, called Recsys, presents a method for recommending electronic commerce with personalized promotions. This expands the scope of recommendation strategies (Zhao et al., 2015).

The Journal of Visual Communication and Image Representation is notable for its contributions to recommender systems and artificial intelligence techniques in electronic commerce. It analyzed personalized recommendations in electronic commerce through clustering, which generated different perspectives on the effectiveness of this approach. Similarly, Applied Soft Computing has a study on a contextual recommendation system for e-commerce. This study helps us understand how to consider context in recommendations. Taken together, these journals have helped develop knowledge in recommender systems and AI techniques in e-commerce.

China, India, and the United States have led the way in scientific research on recommender systems and AI techniques in e-commerce. China has done studies on hybrid recommendation systems for e-commerce websites (Wenxing, Yang, Shunzhi & Maoqing, 2010) and deep learning-based advertising recommendations (Zhou, 2020). India has also made contributions to this field. Patil and Rao (2019) explored the role of AI in e-commerce website design, and Addagarla and Amalanathan (2020) presented a research approach. In the United States, Agichtein and his team held a workshop at the WSDM 2020 conference on conversational systems for e-commerce recommendations and searches (Agichtein, Hakkani-Tür, Kallumadi & Malmasi, 2020).

Other countries have also made important contributions. In Sweden, Brodén et al. (2018) did a pilot study on recommendations in e-commerce. In Spain, Vanesa Aciar, Serarols-Tarres, Royo-Vela and Esteva (2007) worked on improving e-commerce through recommendations using intelligent agents. Indonesia has also made contributions to the field, with research by Pujastuti et al. (2022) on recommender systems for sparse evaluation matrices in e-commerce. South Korea has also been a leader in this area, developing a contextual recommendation system for e-commerce with Khan, Hussain, Iltaf, Kim and Jeon (2021). These countries have helped to improve knowledge and research in recommender systems and AI techniques.

It's important to recognize the impact of contributions that aren't mentioned in the results but are important in the field. Song, Wu, Song and Stojanovic (2023) study how to estimate states in reaction-diffusion neural networks under attack. This helps us understand how to keep neural systems safe. Similarly, Zhang, Song, Sun and Stojanovic (2023) present a hybrid approach based on fuzzy security for secure filtering in nonlinear parabolic partial differential equation systems affected by cyber attacks.

Also, although they don't focus on recommendation systems in e-commerce, works like Wei, Huang and Fu (2007) give a general overview of recommendation systems in e-commerce, while Schafer, Konstan and Riedl (1999) present a pioneering perspective on recommendation systems in e-commerce, establishing important foundations in the field. These contributions are essential to understanding e-commerce and recommender systems.

This literature review looks at how bibliometrics relate to recommendation systems and artificial intelligence in e-commerce. It also looks at where different types of populations have been studied to understand the topic's scientific distribution. Fig. 9 shows where this research was done.

The findings show that the theme has been most validated in Asia and Europe. Many Asian countries, including Turkey, Israel, India, China, South Korea, and Japan, have led the way in studying recommendation systems and artificial intelligence in the context of electronic commerce. Similarly, in Europe, countries like Sweden, Spain, Portugal, the Netherlands, Belgium, Germany, and Italy have also addressed this issue. These are just a few examples. The research covers many more countries.

4.3. Analysis of the thematic evolution of recommender systems and AI techniques in e-commerce

In the early days of research on recommender systems and AI techniques in e-commerce, the idea of "multiagent systems" was a big part of the development (see Fig. 6). The work "Increasing effectiveness in e-commerce: recommendations applying intelligent agents" was a key moment in understanding and using intelligent agents in e-commerce. It looked at how collaborative and autonomous systems could help online shops make better product and service recommendations (Vanesa Aciar

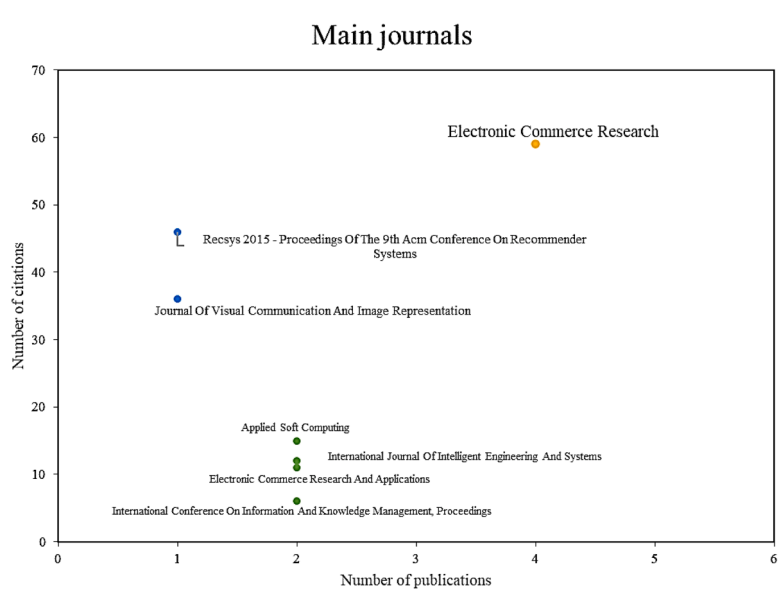

Fig. 4. Main journals. Compiled from Scopus and Web of Science.

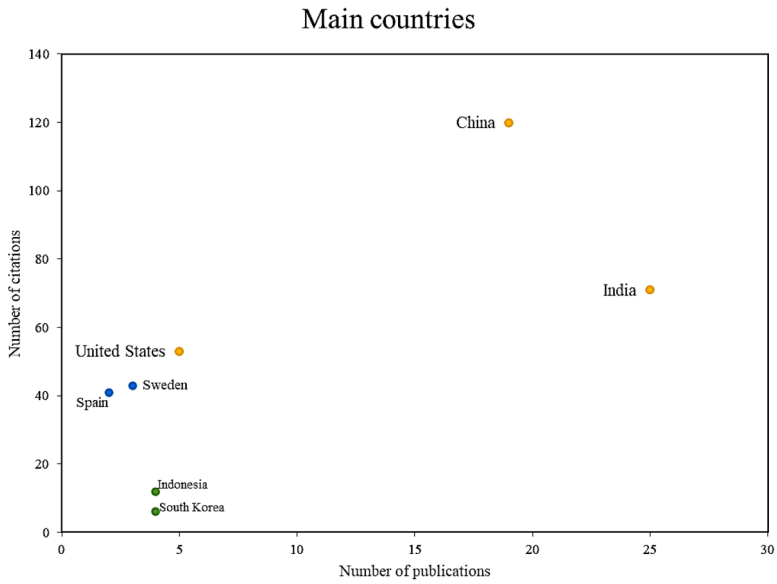

Fig. 5. Main Countries. Own elaboration from Scopus and Web of Science.

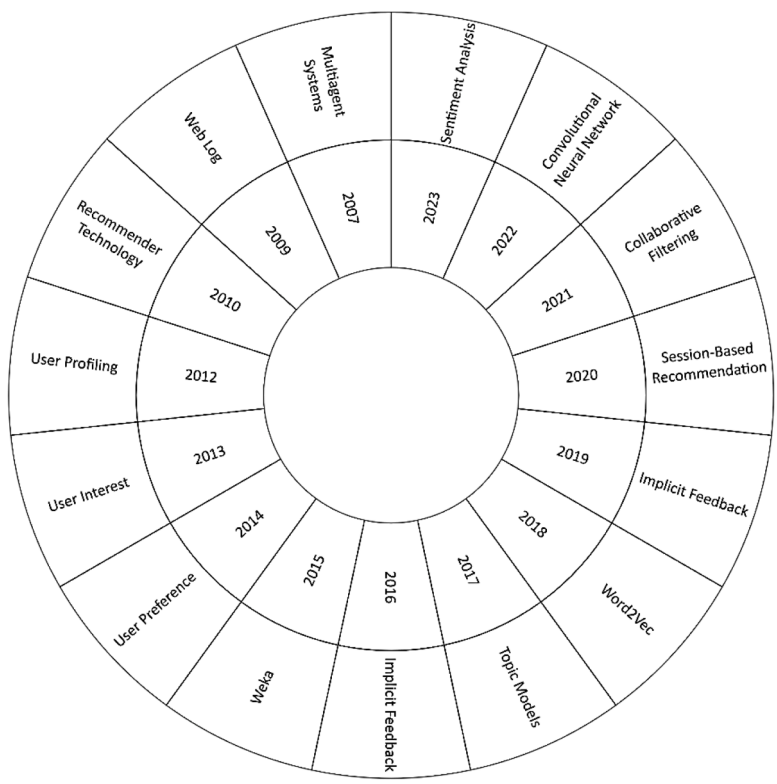

Fig. 6. Thematic evolution. Compiled from Scopus and Web of Science.

et al., 2007). These early works laid the groundwork for future developments in the field. They expanded our understanding of topics like sentiment analysis, convolutional neural networks, collaborative filtering, session-based recommendation, and implicit feedback, which are now used in e-commerce.

In 2019, we focused on "implicit feedback" because it helped us find ways to improve recommendations in e-commerce. In 2020, session-based recommendations were important. This year, a framework for exploring and exploiting diverse recommendation components was explored, emphasizing adapting to changing user preferences in online browsing sessions (Brodén, Hammar, Nilsson & Paraschakis, 2020).

Another important concept is "collaborative filtering." This was studied a lot in 2021. A machine learning system was created that uses collaborative filtering to make e-commerce recommendations more personal. This shows how users can help each other to make better recommendations (Anitha & Kalaiarasu, 2021). In 2022, researchers focused on "convolutional neural networks" (CNNs). They developed a content-based recommendation system for online stores that used CNNs to improve recommendations. This research showed how computer vision can be used in this context (De, Banerjee, Rath, Swain & Samant, 2022).

In e-commerce, several concepts have emerged as important areas of

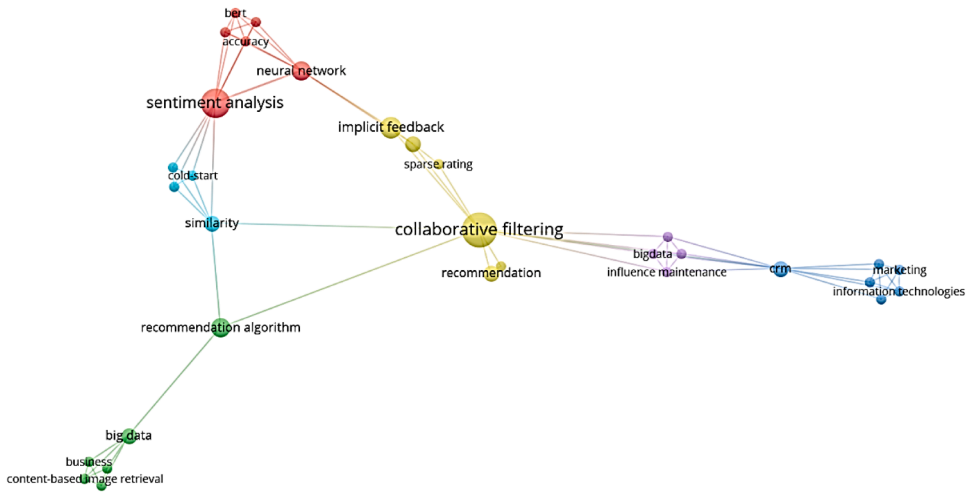

Fig. 7. Keyword co-ocurrence network. Own elaboration from Scopus and Web of Science.

research. One of these is "sentiment analysis," which has become very important in 2023. Since that year, researchers have studied how to combine different types of analysis to create better recommendations for customers in e-commerce (Karn et al., 2023).

These concepts are an important step forward in the development of recommender systems and AI techniques in e-commerce. They offer new ways to make online shopping better.

4.4. Analysis of the thematic clusters on recommender systems and AI techniques in e-commerce

The main keyword co-occurrence network shows different groups of related terms in the scientific literature on recommender systems and AI techniques in e-commerce. The yellow cluster is notable for its terms, including "collaborative filtering," "sparse rating," "implicit feedback," and "recommendation." This cluster is about improving e-commerce recommendations. This is shown in the work of Paraschakis et al. (2015), who compared top-n recommenders from an industry perspective. Hanafi (2022) suggested using different techniques to improve e-commerce recommender systems. Mixing SDAE, attention mechanisms, and matrix factorization helps e-commerce recommender systems give more accurate ratings. Kottage, Jayathilake, Chankuma, Ganegoda and Sandanayake (2018) developed a preference-based recommendation system for clothing e-commerce sites.

The second most relevant cluster, in red, includes keywords like "sentiment analysis," "neural network," "accuracy," and "Bert." This cluster shows how sentiment analysis and neural networks can improve e-commerce recommendations. It also shows how users' feelings affect personalised recommendations (Karn et al., 2023).

Thematic clusters show what researchers are interested in and how different ideas in recommender systems and AI techniques in e-commerce connect.

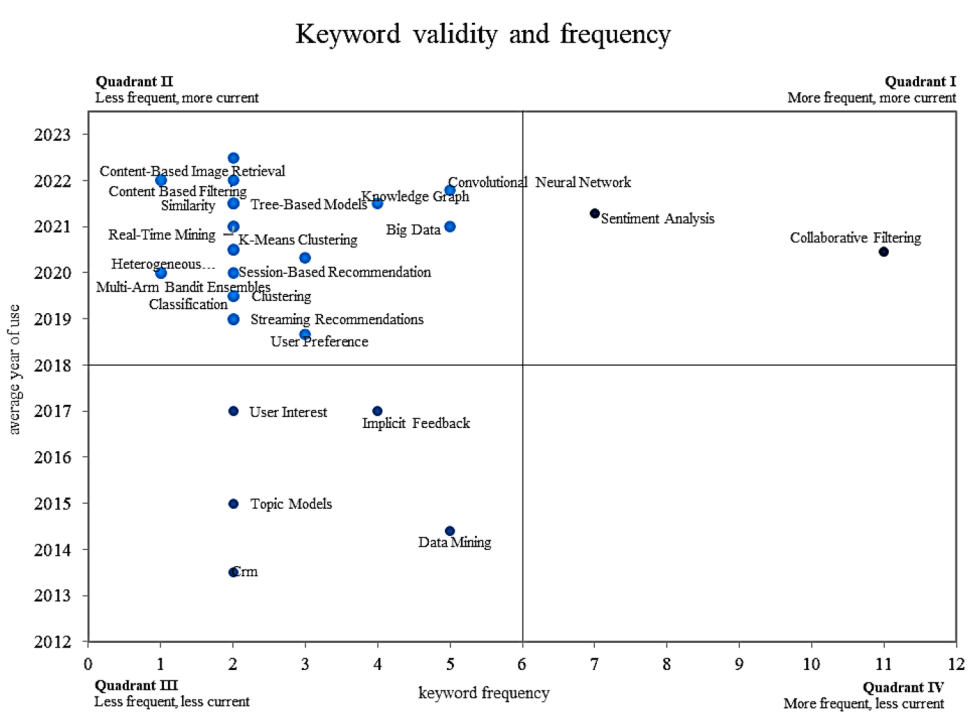

Fig. 8. Validity and frequency of keywords. Own elaboration from Scopus and Web of Science.

4.5. Analysis of the frequency and conceptual validity around recommender systems and AI techniques in electronic commerce

The Cartesian plane in the results section shows how recommender systems and AI techniques in electronic commerce are developing. Keywords in this quadrant are "content-based image retrieval" and "content-based filtering" (see Fig. 8). These terms show that more and more people are interested in using image-based content filtering and retrieval techniques in e-commerce. These concepts are becoming more important because users want personalized recommendations that look good.

Content-based image retrieval helps e-commerce sites make better recommendations by combining visual and content information about products. Images are important for choosing products and having a good experience (Bagwari, Sinha, Singh, Garg & Kanti, 2022). Content-based filtering looks at the content of items to generate personalized recommendations. In e-commerce, it helps make recommendations more relevant by looking closely at the content of products (Tantyo & Mauritsius, 2022). These new ideas could help improve accuracy and personalization in e-commerce recommender systems.

Quadrant 1 shows the concepts that are growing fast and are at the forefront of e-commerce recommender system research and AI techniques. These include "sentiment analysis" and "collaborative filtering". These concepts are still important because they help make e-commerce recommendations more personal and accurate.

Sentiment analysis helps businesses understand what customers think about their products and services. This helps them make better recommendations. Some examples of this are customer-centric hybrid recommendation systems that integrate sentiment analysis. These have been shown to improve the user experience in e-commerce. On the other hand, problems and solutions in recommendation systems enabled by deep learning have also been explored, showing how sentiment analysis helps optimize such systems (Almahmood & Tekerek, 2022).

Collaborative filtering is a reliable way to make recommendations. This approach uses past user-product interactions to provide relevant, personalized recommendations. This method improves recommendations by using information on user behavior. Some authors have created recommender systems based on machine learning that use collaborative filtering to improve recommendations. This helps users and improves the online experience (Anitha & Kalaiarasu, 2021). Other authors have studied how collaborative recommendation approaches can be used in the e-commerce real estate market.

4.6. Classification of keywords related to recommender systems and AI techniques in e-commerce according to their function

Table 1 shows the most common and fastest-changing keywords in recommender systems and AI techniques in e-commerce. They are grouped by their function. This classification helps us understand the main characteristics and uses of each category, giving us a good overview of the latest developments in this field.

4.7. Role and literature of optimization algorithms

AI has helped e-commerce by improving recommendations and decision-making. These algorithms, inspired by biology and nature, have shown they can solve complex problems and optimize systems in

Table 1
Classification of keywords according to their function. Own elaboration from Scopus and Web of Science.

Keyword	Associated Tools	Applications	Characteristics
Content-Based Image Retrieval	Visual Feature Extraction, Image Analysis	Product Recommendation	Use visual features
Content-Based Filtering	User Profiling, Text Analysis	Personalized Content Recommendation	Based on user preferences
Convolutional Neural Network	Deep Learning, Image Processing	Image Recognition, Content Analysis	Image recognition and analysis
Knowledge Graph	Graph Database, Semantic Modeling	Knowledge Representation, Search	Structured knowledge representation
Sentiment Analysis	NLP, Machine Learning	Customer Review Analysis, Feedback	Evaluate emotions in text
Collaborative Filtering	User-Item Interaction, Matrix Factorization	Recommendation, Group Decision-Making	Based on user behavior

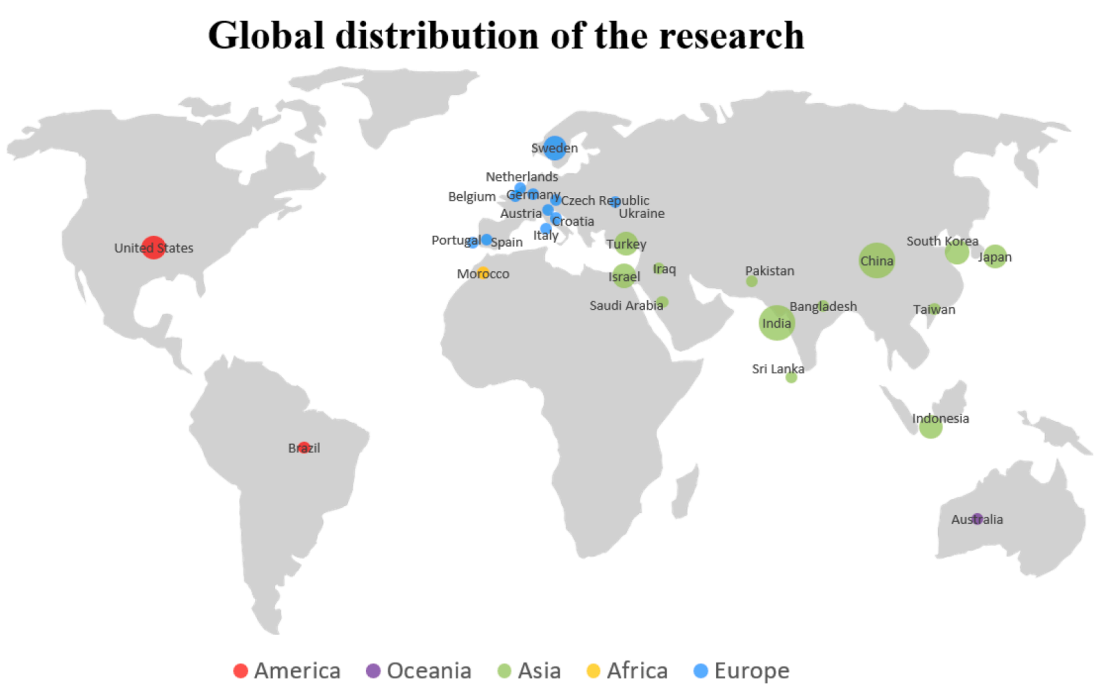

Fig. 9. Global distribution of the research. Own elaboration from Scopus and Web of Science.

changing environments. One of the most important methods is the gravitational search algorithm, which shows how celestial bodies move under the influence of gravity. This method has been used to improve recommendation systems, making them more accurate and efficient (Kourepinis, Iliopoulou, Tassopoulos & Beligiannis, 2024; Somu, MR, Kaveri, Krithivasan & VS, 2020).

The literature shows that optimizing the system of inclined planes is a promising strategy in e-commerce. This approach has been used to improve pricing and online purchases. These methods help solve complex e-commerce problems. They use natural principles and algorithms to improve recommendation systems (Mozaffari, Abdy & Zahiri, 2016; Sarkar et al., 2023).

The latest research shows that these methods can improve e-commerce profits and efficiency. For example, Mozaffari et al. (2016) show how the inclined plane optimization (IPO) system can help solve optimization problems in many areas, including e-commerce. Khamees, El-Rafei, Badra and Abdelaziz (2017) also look at how evolutionary algorithms can help solve problems in electronic commerce.

4.8. Practical implications

This analysis offers insights and guidance for researchers, professionals, and companies in this evolving field. The shift from studying "Multiagent Systems" to analyzing "Sentiment Analysis," "Convolutional Neural Networks," "Collaborative Filtering," "Session-Based Recommendation," and "Implicit Feedback" reflects a change in how researchers approach these topics, which have been adapted to the needs of the current market. This shift suggests that researchers and practitioners should consider new techniques to improve e-commerce recommender systems.

The main topic cluster around terms like "collaborative filtering," "sparse rating," "implicit feedback," and "recommendation" shows how these concepts are connected. This provides a good path for further research and development. This grouping suggests that focusing on user collaboration, handling implicit feedback, and improving recommendations is important. E-commerce organizations and professionals can benefit from using these strategies to improve the user experience and customer loyalty.

The keyword analysis shows that new ideas are becoming more popular. These include "Content-Based Image Retrieval," "Convolutional Neural Network," and "Knowledge Graph." This shows that image analysis and knowledge graphs are becoming more important. Ecommerce companies can use these new approaches to give customers better, more personalised recommendations. This will make customers happier and more likely to buy. At the same time, more focus on sentiment analysis and collaborative filtering shows how important it is to understand what customers want and how they feel. This will help companies market their products better and make them more personalised.

Because customer preferences and market trends change all the time, ecommerce companies need to keep learning and adapting their recommendation systems. This study shows that keeping recommendation models up to date is important. Professionals and companies should think about ways to get new data and adapt models to detect changes in user behavior and product offerings. Updating approaches can help keep recommendations relevant and accurate over time.

Artificial intelligence is changing how businesses and consumers buy things online. This means we must think about the challenges and opportunities in terms of technology, money, and ethics. Academics also face new challenges in this new paradigm (Krishna et al., 2023). Personalized recommendation systems are also affecting e-commerce. Molaie and Lee (2022) say these systems can affect sales diversity. Conversely, content filtering can expose users to niche products and make sales more even. This shows how important it is to design recommendation systems to influence sales and the e-commerce economy.

Recommendation systems affect how consumers buy things. It's important to make these systems fair and honest. Your study shows that people are using more complex techniques like sentiment analysis and collaborative filtering. This means we need to understand and deal with any bias or discrimination in the recommendations. Algorithms should be transparent, understandable, and fair to users. They should also provide control and explanation of how recommendations are generated.

New techniques like "session-based recommendation" and "implicit feedback" show that companies are trying to make online recommendations more personal. This can help customers find what they want more easily and make them happier. This trend also shows the importance of having a technology that can process and respond to user interactions in real time. This helps to create a personalized experience.

This bibliometric analysis also shows ways that recommender systems in e-commerce can be improved. New techniques like image retrieval and content-based filtering suggest a promising approach to recommendation personalization. This goes beyond considering text. Integrating images and other content into recommendation algorithms can enhance the user experience and improve recommendations. This is especially useful in sectors like fashion, interior design, and gastronomy, where appearance and style are important.

The study shows that it is important to work together in different fields to create better recommendation systems. The combination of sentiment analysis, artificial intelligence, and consumer psychology can help us understand how people think and act. There has been little research on how to use psychological data in these algorithms, such as how consumers see themselves (Feng, 2023). This can help companies understand what customers want and how they act in online stores. Companies can work with experts to create better recommendation systems. These systems should consider what customers say and what they really want.

4.9. The practical applications of this theory in real-world settings

Two new techniques let websites change recommendations based on what users do online. This is especially useful in e-commerce, where suggestions can affect purchases. For example, if someone searches for summer clothes, they might get suggestions for similar items based on what they've been looking at. This makes the customer experience better and increases the chance of them buying.

Integrating sentiment analysis with collaborative filtering helps businesses understand customers better, which is good for personalizing shopping experiences and making better marketing decisions. An online store can analyze product reviews to identify positive or negative sentiments and adjust its recommendations and promotions. This helps companies understand what customers want and need, which makes customers happier and more loyal.

This is useful in fashion, interior design, and gastronomy, where images are important. A fashion app could let users upload images of clothes they like and get recommendations of similar items in the store. Adding visual content to recommendation systems makes suggestions more accurate and makes shopping easier for users.

Recommender systems can use knowledge graphs to suggest products based on more than just what a user has bought. For example, an online bookstore could suggest books based on topics, favorite authors, and reviews from other readers with similar interests. This makes recommendations more personal and relevant, which can increase sales and customer satisfaction.

4.10. Limitations

The current research on recommender systems and AI techniques in e-commerce has some limitations. First, the databases used, like Scopus and Web of Science, might have left out other useful sources of information, which could have made the data incomplete. Also, the search

terms could have missed some important contributions because the quality and precision of the search terms could have been poor.

The current bibliometric analysis has focused on two databases, Scopus and Web of Science. This may have left out other sources of information. The two databases may not have included all research on recommender systems and AI in e-commerce. Future research should include more databases. To get a fuller picture of the research in this fast-changing field, it's best to include both academic and industrial sources. This could reveal new trends, approaches, and contributions that have not been captured, enriching global understanding and providing a solid foundation for decision-making in research and development in the field of electronic commerce.

Also, when using Excel and VOSviewer to analyze bibliometric data, there is a chance of human error. Even though we try to avoid mistakes, computers can make them. The results could be biased because researchers decide what to include and how to interpret the data.

Another limitation is that the research field of recommender systems and AI techniques in e-commerce is changing and dynamic. Some recent developments and advances may not have been captured in the data, which could affect the results. The present bibliometric analysis must be interpreted with caution because of its limitations. This is to obtain a more complete and accurate understanding of the research landscape in this field.

Also, the research doesn't have a more detailed statistical analysis or mathematical modeling to show trends and correlations. While tools like Microsoft Excel® and VOSviewer® were used to show how the data is related, they don't show how the data changes over time. The research can't identify patterns and relationships between the variables if there's no more rigorous statistical analysis. Future studies should use more advanced statistical techniques to understand the research landscape in this field better.

The present bibliometric analysis covers a wide time period without focusing on the most recent developments. Future studies could benefit from a more rigorous investigation of recommender systems and artificial intelligence techniques. The last five years of literature would show the most current and emerging trends. This would provide a more up-to-date and relevant perspective on recent advances and innovations in electronic commerce. Also, focusing on the recent period helps identify technologies and methods that are becoming more common in the industry. This would help future research and development.

4.11. Research gaps

Table 2 lists the main problems in recommender systems and AI techniques in e-commerce. These are important areas for future research. These deficiencies result from an analysis of the literature and show where current research has not yet provided satisfactory answers or where there are opportunities for advancing knowledge. The table shows what needs to be done and why. This approach helps researchers find promising areas of study and a plan for future research in recommender systems and AI techniques in e-commerce.

4.12. Research agenda

This study proposes a research agenda for AI recommender systems in e-commerce. It includes the main, growing, and emerging keywords from the scientific literature on the topic. Fig. 10 shows the current state of knowledge in this area.

The tool lets you study patterns, users, trends, and hidden relationships in large amounts of data. Future research should study how to use advanced data mining to better understand how users interact with e-commerce platforms. Also, combining data mining with other AI methods, like neural networks and deep learning, could create more intelligent and personalized recommendation systems. It would also be good to look at how data mining can help to stop people from cheating and to make online shopping safer.

Sentiment analysis is a key tool in e-commerce recommendation systems and AI techniques. This technique is used to identify user preferences and to detect satisfaction or dissatisfaction with e-commerce platforms. Future studies should look at how to improve sentiment analysis and how it can be used in recommendation algorithms to make more intelligent systems that take into account the emotional side of purchases. It would also be useful to see if sentiment analysis can be used in different cultures and languages to make recommendations that work for everyone. Finally, we should look into how sentiment analysis can help users and e-commerce platforms interact better through chatbots that understand customers.

The field of recommender systems and AI techniques in e-commerce has been the subject of interest in recent times due to the potential of Recurrent Neural Networks (RNNs) to model data streams. Currently, RNNs are being used to predict purchase patterns and user preferences based on their history, which enables the selection of more relevant and timely recommendations. Future research could look at ways to make RNNs faster and more efficient, especially in e-commerce. New RNN architectures like LSTM and GRU could help with long-term temporal dependencies. It would also be useful to see how RNNs could be used to create product descriptions based on user behavior to improve online shopping.

Collaborative filtering is still important in e-commerce. This is based on the idea that similar users' preferences and behaviours can be used to make recommendations to a particular user. Currently, collaborative filtering is being improved by new approaches based on matrices and factorization models. However, future research could make this technique better by finding new ways to deal with the "cold start" problem and data in large systems.

The rise of big data has changed how e-commerce uses recommender systems and AI. Currently, big data is used to track customer interactions in real time, allowing recommendations to be changed quickly. Future studies could improve this by looking at how to use big data tools like Apache Hadoop and Spark to make recommender systems faster and

Table 2
Research gaps. Compiled from Scopus and Web of Science.

Category	Investigative Gaps	Justification	Future Research Questions
Thematic Gaps	1. Use AI and ethics in recommendations. 2. How AI is used in e-commerce.	Ethics in AI is important because it affects society and the economy. New e-commerce techniques need to be studied.	How can ethical principles be added to recommendation algorithms? How can AI improve e-commerce personalization?
Geographic gaps	1. Research on recommender systems in emerging markets. 2. Research on electronic commerce between countries.	Most research is on developed countries, so we don't know much about developing markets (see Fig. 9). International collaboration can help.	How can recommender systems be adapted to emerging markets? How can we work together to improve e-commerce research?
Interdisciplinary Gaps	1. Personalize recommendations with psychological approaches. 2. Marketing methods for recommendations.	Personalization could benefit from psychology. Marketing can make recommendations more effective.	How can consumer decision-making theories help with personalization? How can traditional marketing help improve user satisfaction with recommendation systems?
Temporary Gaps	1. How have recommender systems been used in e-commerce? 2. Study of how AI will affect e-commerce.	Long-term research can show how things change over time. AI is changing fast. We need to look ahead.	How have recommender systems changed over time? What AI techniques will impact e-commerce in the future?

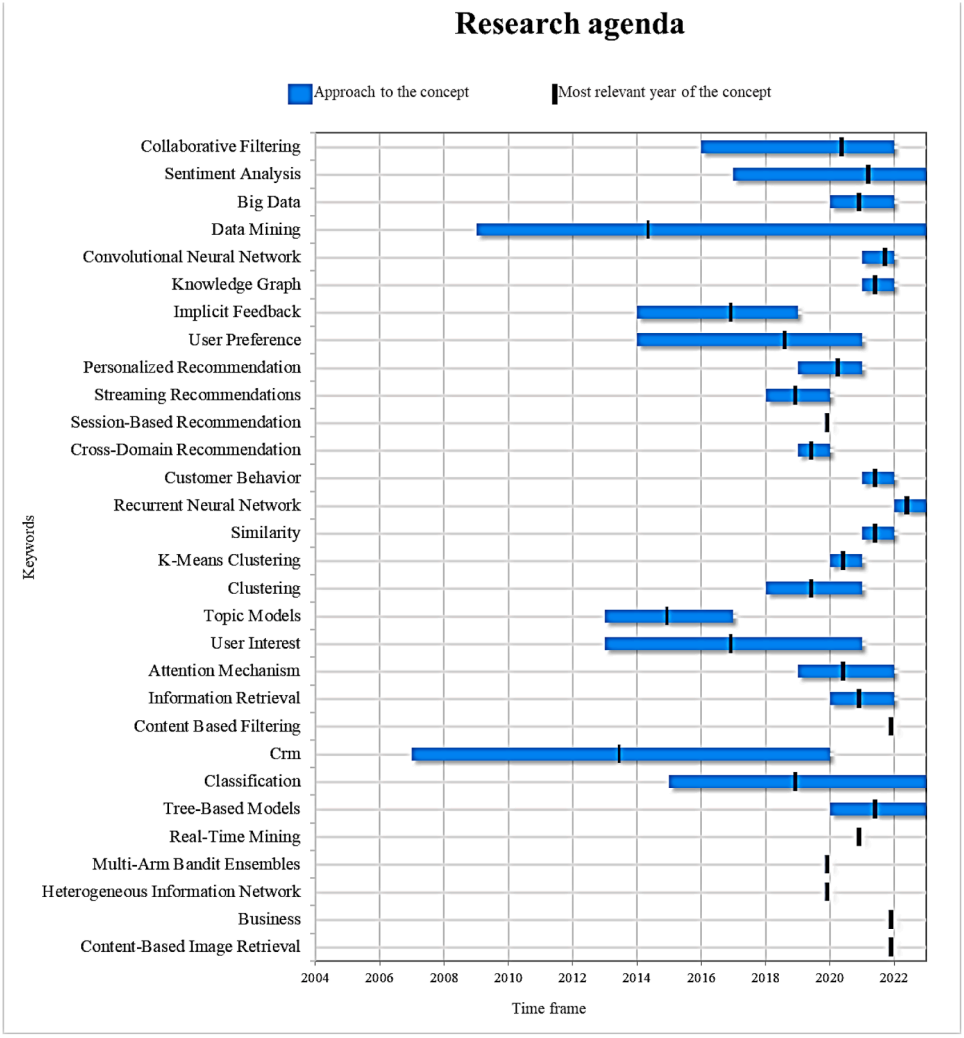

Fig. 10. Research agenda. Own elaboration from Scopus and Web of Science.

more efficient. Also, studying more advanced analytical techniques like stream mining and graph analysis could help identify more useful patterns in e-commerce data.

Convolutional neural networks (CNNs) are important in recommender systems and AI techniques in e-commerce because they can process and extract features from unstructured data. Currently, CNNs are being used to improve recommendations by incorporating images and user feedback. Future research could explore how to adapt and optimize CNNs for real-time e-commerce data processing. Also, combining CNNs with other recommendation methods could create more powerful and versatile hybrid systems. It would also be good to look at how CNNs can automatically identify features in product images, which would help make products more personalised and attractive.

Knowledge graphs are important in e-commerce and AI. They are used to make recommendations more relevant by considering relationships between products and users. Future research could look into building and maintaining knowledge graphs in real time. Also, we could combine knowledge graphs with other recommendation techniques to create more informed and contextual systems.

5. Conclusions

This bibliometric study found that 2021 and 2022 were the most interesting years for this subject. This shows that research in this field is constantly changing and growing.

The authors Paraschakis D and Nilsson BJ, the journal Electronic Commerce Research, and countries such as China, India, and the United States have been identified as the primary research references. This overview shows the main people and sources in research on recommender systems and AI in e-commerce.

The focus of scientific research has changed a lot. At first, researchers focused on Multiagent Systems. Now, they study concepts like Sentiment Analysis, Convolutional Neural Networks, Collaborative Filtering, Session-Based Recommendation, and Implicit Feedback. The research community can adapt its approaches to new technology and e-commerce needs. The analysis shows that keywords are related. The cluster formed by "Collaborative Filtering," "Sparse Rating," "Implicit Feedback," and "Recommendation" is particularly strong. This shows that the research areas are connected.

Keywords related to emerging and growing topics include sentiment analysis and collaborative filtering. Other terms to watch are content-based image retrieval, content-based filtering, convolutional neural network, and knowledge graph. These topics offer promising research avenues for understanding how new trends influence recommender systems in e-commerce.

Finally, this study has looked at how recommender systems and AI techniques in e-commerce have changed over time. The results can help future researchers understand more about the issues in this field and how to improve it.

Author contributions

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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